

Ignacio Magaña Hernandez

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Employment

Carnegie Mellon University <i>McWilliams Postdoctoral Fellow</i>	Pittsburgh, PA <i>June 2023 - Present</i>
University of Wisconsin-Milwaukee <i>NSF Graduate Research Fellow</i>	Milwaukee, WI <i>September, 2017 - May 2023</i>
The Chinese University of Hong Kong <i>Undergraduate Research Assistant</i>	Hong Kong, Hong Kong <i>March, 2016 - January 2017</i>
Caltech <i>LIGO SURF Fellow</i>	Pasadena, CA <i>June, 2015 - September, 2015</i>

Education

University of Wisconsin-Milwaukee <i>PhD in Physics</i>	Milwaukee, WI <i>September 2017 - May 2023</i>
University of California, Santa Barbara <i>B.S., Physics (Highest Honors)</i>	Santa Barbara, CA <i>September 2014 - September 2017</i>
Foothill College <i>Transfer Student, Physics</i>	Los Altos Hills, CA <i>September 2012 - June 2014</i>

Awards & Fellowships

• CMU McWilliams Postdoctoral Fellowship <i>McWilliams Center for Cosmology and Astrophysics three year fellowship.</i>	2023 - Present
• CMU McWilliams Seed Grant <i>Internal grant funding exploratory projects to generate preliminary results for high impact research.</i>	2025
• UWM Advanced Opportunity Program Fellowship <i>UWM fellowship supporting underrepresented graduate students for three years.</i>	2019 - 2023
• NSF Graduate Research Fellowship <i>NSF fellowship supporting graduate students for three years.</i>	2017 - 2023
• NASA Wisconsin Space Grant Consortium Fellowship <i>NASA funded fellowship for graduate research in STEM at Wisconsin institutions.</i>	Summer 2020, 2021, 2022
• UWM Physics Department Lichtman Fellowship <i>Departmental fellowship awarded for exceptional academic achievement in physics.</i>	Spring 2020
• UCSB Physics Department Highest Academic Honors <i>Awarded to graduating physics students with at least a 3.9 GPA in physics courses only.</i>	Summer 2017
• LIGO SURF Blanco Fellowship <i>Summer Research Fellowship at Caltech/LIGO, supporting undergraduates in GW science.</i>	Summer 2015
• UCSB Dean's list <i>Recognition for maintaining a GPA of 3.75 per quarter.</i>	Fall 2014 - Spring 2017
• NSF Science, Math and Engineering (S-STEM) Scholarship <i>NSF scholarship providing support to students pursuing STEM degrees.</i>	Fall 2013 - Spring 2014

Publication Summary

h-index: 52 / 17 Publications: 81 / 35 Citations: 44k / 2.0k

First number includes LIGO collaboration papers; second number excludes them
Citations according to inspirehep.net in September 2026

First/Second Author Publications

1. *Dark siren cosmology with the Dark Energy Spectroscopic Instrument*
I. Magaña Hernandez, A. Palmese et al.
Under DESI internal review

2. *Data-Driven Trends and Subpopulations in the Gravitational Wave Binary Black Hole Merger Population with UMAP*
A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and J. Gassert
([arXiv:2603.06566](#), 0 citations)
3. *Lucky Strikes: On the Origins of GW190814 Through Isolated Binary Evolution*
I. Magaña Hernandez and K. Breivik
([arXiv:2511.16648](#), 1 citations) Submitted to *ApJL*
4. *Spectral siren cosmology from gravitational-wave observations in GWTC-4.0*
I. Magaña Hernandez and A. Palmese
([arXiv:2509.03607](#), 3 citations) Submitted to *Physical Review Letters*
5. *Astrophysics informed Gaussian processes for gravitational-wave populations: Evidence for the onset of the pair-instability supernova mass gap*
I. Magaña Hernandez and A. Palmese
([arXiv:2508.19208](#), 7 citations) Submitted to *Physical Review Letters*
6. *A new bump in the night: evidence of a new feature in the binary black hole mass distribution at $70 M_{\odot}$ from gravitational-wave observations*
I. Magaña Hernandez and A. Palmese
Phys. Rev. D 111, 083031 (2025) ([arXiv:2407.02460](#), 10 citations)
7. *Searching for binary black hole sub-populations in gravitational wave data using binned Gaussian processes*
A. Ray, **I. Magaña Hernandez**, K. Breivik, J.D.E. Creighton
ApJ 991 17 (2025) ([arXiv:2404.03166](#), 31 citations)
8. *Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models*
I. Magaña Hernandez and A. Ray
([arXiv:2404.02522](#), 24 citations) Submitted to *Physical Review D*
9. *On the association of GW190425 with its potential electromagnetic counterpart FRB 20190425A*
I. Magaña Hernandez, V. D’Emilio, S. Morisaki, M. Bhardwaj and A. Palmese
ApJL 971 L5 (2024) ([arXiv:2404.02496](#), 13 citations)
10. *Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes*
A Ray, **I. Magaña Hernandez**, S. Mohite, J.D.E. Creighton and S. Kapadia
ApJ 957 37 (2023) ([arXiv:2304.08046](#), 66 citations)
11. *Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers*
I. Magaña Hernandez
([arXiv:2211.01272](#), 11 citations) Submitted to *Physical Review D*
12. *Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers*
I. Magaña Hernandez
Phys. Rev. D 107, 084033 (2023) ([arXiv:2112.07650](#), 19 citations)
13. *A Bayesian statistical framework for identifying strongly-lensed gravitational-wave signals*
R.K.L. Lo and **I. Magaña Hernandez**
Phys. Rev. D 107, 123015 (2023) ([arXiv:2104.09339](#), 78 citations)
14. *Identifying strong gravitational-wave lensing during the second observing run of Advanced LIGO and Advanced Virgo*
X. Liu, **I. Magaña Hernandez** and J.D.E. Creighton
ApJ 908 97 (2021) ([arXiv:2009.06539](#), 89 citations)
15. *Cosmological inference using gravitational wave standard sirens: A mock data analysis*
R. Gray, **I. Magaña Hernandez** et al.
Phys. Rev. D 101, 122001 (2020) ([arXiv:1908.06050](#), 211 citations)

Other Short Author Work

1. *One Year and One Night to 1% in H_0 : Efficient Spectroscopic Strategy for Dark Siren Cosmology*
Y. Dang, A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and B. S. Sathyaprakash
([arXiv:2607.17284](#), 0 citations)
2. *Electromagnetic Follow-up of the Sub-Solar Mass Gravitational Wave Candidate S251112cm: Kilonova Constraints and a Coincident IIb Supernova*
X. J. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2605.10940](#), 0 citations)
3. *Wide Jets or Low Rates: Reconciling Short GRB and Gravitational-Wave Neutron Star Merger Rates*

- K. Kunnumkai, A. Palmese, B. O'Connor, A. Farah and **I. Magaña Hernandez**
([arXiv:2604.05046](#), 0 citations)
4. *Probing the environment around GW170817 with DESI: insights on galaxy group peculiar velocities for standard siren measurements*
A. J. Amsellem et al. (including **I. Magaña Hernandez**)
([arXiv:2512.08818](#), 0 citations)
 5. *AT2025ulz and S250818k: Investigating early time observations of a subsolar mass gravitational-wave binary neutron star merger candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2510.24620](#), 1 citations)
 6. *AT2025ulz and S250818k: Deep X-ray and radio limits on off-axis afterglow emission and prospects for future discovery*
B. O'Connor et al. (including **I. Magaña Hernandez**)
([arXiv:2510.23728](#), 1 citations)
 7. *AT2025ulz and S250818k: Leveraging DESI spectroscopy in the hunt for a kilonova associated with a sub-solar mass gravitational wave candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2510.23723](#), 1 citations)
 8. *Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations*
L. Hu et al. (including **I. Magaña Hernandez**)
ApJL 990 L46 (2025) ([arXiv:2506.22626](#), 3 citations)
 9. *A Dark Siren Measurement of the Hubble Constant with the LIGO/Virgo Gravitational Wave Event GW190412 and DESI Galaxies*
W. Ballard, A. Palmese, **I. Magaña Hernandez** et al.
Res. Notes AAS 7 250 (2023) ([arXiv:2311.13062](#), 22 citations)
 10. *Follow-up analyses to the O3 LIGO–Virgo–KAGRA lensing searches*
J. Janquart et al. (including **I. Magaña Hernandez**)
MNRAS 526, 3832 (2023) ([arXiv:2306.03827](#), 55 citations)
 11. *Challenges for Fast Radio Bursts as Multi-Messenger Sources from Binary Neutron Star Mergers*
M. Bhardwaj, A. Palmese, **I. Magaña Hernandez**, V. D’Emilio and S. Morisaki
ApJ 977 122 (2024) ([arXiv:2306.00948](#), 27 citations)
 12. *Localization of binary neutron star mergers with a single Cosmic Explorer*
P. Baral, S. Morisaki, **I. Magaña Hernandez** and J.D.E. Creighton
Phys. Rev. D 108, 043010 (2023) ([arXiv:2304.09889](#), 14 citations)
 13. *The Hitchhiker’s guide to the galaxy catalog approach for gravitational wave cosmology*
J. Gair et al. (including **I. Magaña Hernandez**)
AJ 166 22 (2023) ([arXiv:2212.08694](#), 75 citations)
 14. *A joint ranking statistic for multi-messenger astronomical searches with gravitational waves*
B. Piotrkowski, A. Baylor and **I. Magaña Hernandez**
Class. Quantum Grav. 39 085010 (2022) ([arXiv:2111.12814](#), 8 citations)
 15. *Rapid model comparison of equations of state from gravitational wave observation of binary neutron star coalescences*
S. Ghosh, X. Liu, J.D.E. Creighton, **I. Magaña Hernandez** et al.
Phys. Rev. D 104, 083003 (2021) ([arXiv:2104.08681](#), 23 citations)
 16. *Gravitational wave friction in light of GW170817 and GW190521*
S. Mastroianni, L. Haegel, K. Karathanasis, **I. Magaña Hernandez** and D.A. Steer
JCAP 02 043 (2021) ([arXiv:2010.04047](#), 38 citations)
 17. *Current observations are insufficient to confidently associate the binary black hole merger GW190521 with AGN J124942.3 + 344929*
G. Ashton, K. Ackley, **I. Magaña Hernandez** et al.
Class. Quantum Grav. 38 235004 (2021) ([arXiv:2009.12346](#), 97 citations)
 18. *First measurement of the Hubble parameter from bright binary black hole GW190521*
S. Mukherjee et al. (including **I. Magaña Hernandez**)
([arXiv:2009.14199](#), 63 citations)
 19. *Bayesian inference for compact binary coalescences with BILBY: Validation and application to the first LIGO–Virgo gravitational-wave transient catalogue*
I.M. Romero-Shaw et al. (including **I. Magaña Hernandez**)

MNRAS 499, 3295 (2020) ([arXiv:2006.00714](#), 492 citations)

20. *A statistical standard siren measurement of the Hubble Constant from the LIGO/Virgo gravitational wave compact object merger GW190814 and Dark Energy Survey galaxies*
A. Palmese et al. (including **I. Magaña Hernandez**)
ApJL 900 L33 (2020) ([arXiv:2006.14961](#), 141 citations)
21. *A standard siren measurement of the Hubble Constant from GW170817 without the electromagnetic counterpart*
M. Fishbach, R. Gray, **I. Magaña Hernandez** et al.
ApJL 871 L13 (2019) ([arXiv:1807.05667](#), 252 citations)

LIGO-Virgo Publications to which I highly contributed:

1. *GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJL 896 L44 (2020) ([arXiv:2006.12611](#), 1823 citations) **Led and wrote the Hubble constant analysis**
2. *A Gravitational-wave Measurement of the Hubble Constant Following the Second Observing Run of Advanced LIGO and Virgo*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJ 909 218 (2021) ([arXiv:1908.06060](#), 386 citations) **Paper writing and analysis team member**
3. *First measurement of the Hubble constant from a dark standard siren using the Dark Energy Survey galaxies and the LIGO/Virgo binary-black-hole merger GW170814*
M. Soares-Santos, A. Palmese et al. and LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJL 876 L7 (2019) ([arXiv:1901.01540](#), 320 citations) **Validated analysis using gwcosmo**

Invited Seminars

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| 1. American Museum of Natural History
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | October 2025 |
| 2. Vanderbilt University
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | April 2025 |
| 3. KAGRA Seminar
<i>Binary black hole populations and cosmology</i> | April 2025 |
| 4. Stanford University (KIPAC Seminar)
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | March 2025 |
| 5. Carnegie Mellon University (STAMPS Seminar)
<i>Binary black hole populations and cosmology</i> | January 2025 |
| 6. Montclair State University
<i>Cosmology, Lensing, and Modified Gravity with Gravitational Waves</i> | September 2023 |
| 7. Carnegie Mellon University
<i>Testing general relativity with strongly lensed gravitational waves</i> | September 2023 |
| 8. Cardiff University
<i>Cosmology and modified gravity with binary black holes</i> | September 2022 |
| 9. UC Berkeley
<i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | June 2022 |
| 10. University of Wisconsin-Milwaukee
<i>Cosmology and modified gravity with binary black hole mergers</i> | April 2022 |
| 11. Eotvos Lorand University
<i>Gravitational wave cosmology with large galaxy surveys</i> | September 2019 |
| 12. University of Glasgow
<i>Measuring the Hubble constant with gravitational-wave standard sirens</i> | August 2019 |
| 13. The Chinese University of Hong Kong
<i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | February 2019 |
| 14. Nagoya University
<i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | February 2019 |

Talks at Conferences

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| 1. Gravitational Waves Physics and Astronomy Workshop | December 2025 |
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2. **COSMO25** October 2025
Gravitational-wave dark siren cosmology with DESI
3. **Multimessenger Astronomy in the Era of Foundational AI** August 2025
Trends in the GWTC-3 Binary Black Hole Merger Population with UMAP
4. **Summer DESI Collaboration Meeting** July 2025
Gravitational-wave cosmology with DESI
5. **APS Global Physics Summit** April 2025
Lucky Strikes: Unraveling the asymmetric origins of GW190814 through binary field evolution
6. **The 6th neighborhood workshop** April 2024
Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models
7. **Gravitational-wave populations: what's next?** July 2023
Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes
8. **APS April Meeting** April 2023
Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers
9. **11th Australasian Conference on General Relativity and Gravitation** February 2022
Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers
10. **Midwest Relativity Meeting** October 2021
Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers
11. **Midwest Relativity Meeting** October 2020
On the candidate electromagnetic counterpart of GW190521 and its implications on the Hubble constant
12. **Gravitational Waves Physics and Astronomy Workshop** October 2019
A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo
13. **Multi-Messenger Astrophysics in the Gravitational Wave Era** September 2019
A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo
14. **YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology** February 2019
A standard siren measurement of the Hubble constant from GW170817 without the electromagnetic counterpart
15. **Midwest Relativity Meeting** October 2018
Cosmological inference using gravitational wave standard sirens: A mock data challenge
16. **19th Annual Physical Society of Hong Kong** June 2016
Feedforward Seismic Noise Cancellation at the 40m Prototype Interferometer

Posters

1. **APS Global Physics Summit** April 2025
A new bump in the night: evidence of a new feature in the binary black hole mass distribution at $70 M_{\odot}$ from gravitational-wave observations
2. **Gravitational Waves Physics and Astronomy Workshop** May 2024
Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models
3. **Gravitational Waves Physics and Astronomy Workshop** May 2024
Backpop: Data Driven Population Synthesis inference on Compact Binary Mergers
4. **LVK Meeting: Cardiff University** September 2022
Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers
5. **Merging Visions: Exploring Compact-Object Binaries with Gravity and Light** June 2019
Identifying Strong Gravitational-wave Lensing during the Second Observing Run of Advanced LIGO and Advanced Virgo
6. **Gravitational Waves Physics and Astronomy Workshop** December 2018
Cosmological inference using gravitational wave standard sirens: A mock data analysis

Other Workshops

1. **Multi-messenger Transients from Binary Mergers and Stellar Explosions** August 2024
Aspen Center for Physics
2. **Black Holes and Neutron Stars with Gravitational Waves** October 2019
Yukawa Institute for Theoretical Physics

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| 3. The New Era of Gravitational-Wave Physics and Astrophysics
<i>Kavli Institute for Theoretical Physics</i> | May - July 2019 |
| 4. Asian-Pacific Winter School and Workshop on Gravitation and Cosmology
<i>Yukawa Institute for Theoretical Physics</i> | February 2019 |
| 5. Les Houches Summer School: Gravitational-Wave Astronomy
<i>Ecole de Physique des Houches</i> | July 2018 |
| 6. Summer School on Gravitational-Wave Astronomy
<i>ICTS-TIFR</i> | August 2016 |

Societies/Affiliations

- Dark Energy Spectroscopic Instrument (DESI) member (June 2023 - Present)
- LIGO Scientific Collaboration member (June 2015 - June 2023)
- American Physical Society
- Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
- National Society of Hispanic Physicists

Student Mentoring

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| • Kaitlyn Pak
<i>Undergraduate student - Multi-messenger associations with GWs</i> | Carnegie Mellon University
September 2025 - Present |
| • Ariel Amsellem
<i>PhD student - Standard siren cosmology.</i> | Carnegie Mellon University
September 2023 - Present |
| • Simon Yeung
<i>PhD student - GW population systematics.</i> | University of Wisconsin-Milwaukee
September 2023 - Present |
| • Julia Yoon
<i>Undergraduate student - Dark siren cosmology with GW170817</i> | Carnegie Mellon University
September 2024 - May 2025 |
| • Diego Michel
<i>Undergraduate student - Gaussian processes for GW populations.</i> | Carnegie Mellon University
September 2023 - May 2024 |
| • Anarya Ray
<i>PhD student (co-advised) - Binned Gaussian processes for GW populations.</i> | University of Wisconsin-Milwaukee
September 2021 - June 2024 |
| • Pratyusava Baral
<i>PhD student (co-advised) - Localization of BNS events with 3G GW detectors.</i> | University of Wisconsin-Milwaukee
September 2021 - May 2023 |
| • Brandon Piotrkowski
<i>PhD student (co-advised) - Multi-messenger association statistics.</i> | University of Wisconsin-Milwaukee
September 2019 - February 2020 |

Skills

- **Programming skills:**
 - Proficiency with: Python (NumPy, SciPy, Pandas, Scikit-learn), JAX, C, Mathematica, LaTeX.
 - Familiar with: Stan, C++, MATLAB/Simulink.
 - LVK related software: Bilby, LALSuite, PyCBC, ligo.skymap, gwcosmo, hanabi.
- **Data Analysis:**
 - Bayesian model selection and parameter estimation, Monte Carlo methods.
 - Digital signal processing, machine learning, and basic numerical relativity experience.
- **Languages:** English (fluent), Spanish (fluent), Korean (intermediate), and Japanese (beginner).

References

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